CHAPTER 11

Some Basic Plasma Phenomena

1. ELECTRON PLASMA OSCILLATIONS

One of the fundamental properties of a plasma is its tendency to maintain electric charge neutrality on a macroscopic scale under equilibrium conditions. When this macroscopic charge neutrality is disturbed, such as to temporarily produce a significant imbalance of charge, large Coulomb forces come into play which tend to restore the macroscopic charge neutrality. Since these Coulomb forces cannot be naturally sustained in the plasma, it breaks into high frequency electron plasma oscillations, which enable the plasma to maintain on the average its electric neutrality.

As a simple example, consider a small spherical region inside a plasma and suppose that a perturbation in the form of an excess of negative charge is introduced in this small region. Because of spherical symmetry, the corresponding electric field is radial and points towards the center (Fig. 1),

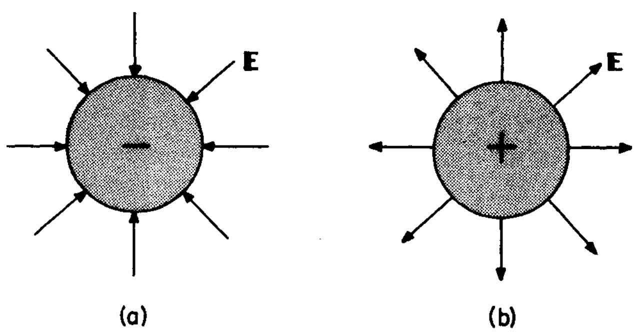

Fig. 1 - The radial electric field $\vec{E}$ produced by a spherical distribution of negative charge (a) forces the electrons to move radially outward, whereas the field produced by a spherical distribution of positive charge (b) forces the electrons to move radially inward.

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forcing the electrons to move radially outward. After a small time interval, since the electrons gain kinetic energy in the course of their motion, more electrons leave the spherical plasma region (due to their inertia) than is necessary to resume the state of electrical neutrality. An excess of positive charge results therefore inside this region and the reversed electric field causes the electrons to move inward now. This movement of electrons inward and outward of the spherical plasma region continues periodically, resulting in electron plasma oscillations. In this way the plasma maintains its macroscopic neutrality on the average, since the total charge inside the spherical region averaged over one period of these oscillations is zero. The frequency of these oscillations is usually very high and since the ions (in view of their mass) are unable to follow the rapidity of these oscillations their motion is often neglected.

To study the characteristics of the electron plasma oscillations we can use the cold plasma model, in which the particle thermal motion and the pressure gradient force are not taken into account. We neglect ion motion and assume a very small electron density perturbation such that

$$n_e(\vec{r},t) = n_0 + n_e'(\vec{r},t) \tag{1.1}$$

where n_0 is a constant number density and $|n_e'| \ll n_0$. Similarly, we assume that the electric field produced, $\vec{E}(\vec{r},t)$, and the average electron velocity, $\vec{u}_e(\vec{r},t)$, are first order perturbations, so that the linearized equations can be used. The linearized continuity and momentum equations become, respectively,

$$\partial n_e'/\partial t + n_0 \, \vec{\nabla} \cdot \vec{u}_e = 0 \tag{1.2}$$

$$\partial \vec{u}_e/\partial t = - (e/m_e) \, \vec{E} \tag{1.3}$$

In the momentum equation we have assumed that the rate of loss of momentum from the electron gas due to collisions is negligible. Considering singly charged ions, the charge density is

$$\rho \, (\vec{r}, t) = - e \, [n_0 + n_e'(\vec{r},t)] + e \, n_0 = - e \, n_e'(\vec{r},t) \tag{1.4}$$

where the ion density was considered to be constant and uniform, and equal to

n_o (neglecting ion motion). Therefore,

$$\vec{\nabla} \cdot \vec{E} = \rho/\varepsilon_o = -(e/\varepsilon_o)\, n_e' \tag{1.5}$$

Eqs. (1.2), (1.3) and (1.5) constitute a complete set of equations to be solved for the variables $n_e'(\vec{r},t)$, $\vec{u}_e(\vec{r},t)$ and $\vec{E}(\vec{r},t)$. Taking the divergence of (1.3) and using (1.2) to substitute for $\vec{\nabla} \cdot \vec{u}_e$, we obtain

$$\partial^2 n_e'/\partial t^2 - (e n_o/m_e)\, \vec{\nabla} \cdot \vec{E} = 0 \tag{1.6}$$

Combining (1.5) and (1.6) to eliminate $\vec{\nabla} \cdot \vec{E}$, yields

$$\partial^2 n_e'/\partial t^2 + \omega_{pe}^2\, n_e' = 0 \tag{1.7}$$

where

$$\omega_{pe} = (n_o\, e^2/m_e\, \varepsilon_o)^{1/2} \tag{1.8}$$

is called the *electron plasma frequency*. (1.7) shows that $n_e'(\vec{r},t)$ varies harmonically in time at the electron plasma frequency,

$$n_e'(\vec{r},t) = n_e'(\vec{r}) \exp(-i\, \omega_{pe}\, t) \tag{1.9}$$

In fact, all first-order perturbations have a harmonic time variation at the plasma frequency ω_{pe}. To justify this statement it is convenient to start with the assumption that all first order quantities vary harmonically in time, as $\exp(-i\, \omega\, t)$. Eqs. (1.2) and (1.3) become, in this case,

$$n_e' = -(i/\omega)\, n_o\, \vec{\nabla} \cdot \vec{u}_e \tag{1.10}$$

$$\vec{u}_e = -(ie/\omega m_e)\, \vec{E} \tag{1.11}$$

which can be combined into

$$n_e' = -(n_o e/\omega^2 m_e)\, \vec{\nabla} \cdot \vec{E} \tag{1.12}$$

Substituting this expression for n_e' into (1.5), yields

$$(1 - \omega_{pe}^2/\omega^2)\ \vec{\nabla} \cdot \vec{E} = 0 \tag{1.13}$$

which shows that a nontrivial solution requires $\omega = \omega_{pe}$. Therefore, all the perturbations vary harmonically in time at the electron plasma frequency. Further, for all variables there is no change in phase from point to point, implying in the absence of wave propagation. The oscillations are therefore *stationary*. Also, (1.11) shows that the electron velocity is in the same direction as the electric field, so that these oscillations are *longitudinal*.

The electron plasma oscillations are also *electrostatic* in character. In order to show this aspect of the oscillations consider Maxwell curl equations with a harmonic time variation,

$$\vec{\nabla} \times \vec{E} = i\ \omega\ \vec{B} \tag{1.14}$$

$$\vec{\nabla} \times \vec{B} = \mu_o\ (\vec{J} - i\ \omega\ \varepsilon_o\ \vec{E}) \tag{1.15}$$

The electric current density is given by

$$\vec{J} = -\ en_o\ \vec{u}_e = (i\ n_o\ e^2/\omega m_e)\ \vec{E} \tag{1.16}$$

where we have used (1.11) for $\vec{u}_e$. Therefore,

$$\vec{\nabla} \times \vec{B} = -\ i\ \omega\ \mu_o\ \varepsilon_o\ \varepsilon_r\ \vec{E} \tag{1.17}$$

where we have defined a relative permittivity by

$$\varepsilon_r = 1 - \omega_{pe}^2/\omega^2 \tag{1.18}$$

For the electron plasma oscillations we have $\omega = \omega_{pe}$, so that $\varepsilon_r = 0$, and (1.17) reduces to

$$\vec{\nabla} \times \vec{B} = 0 \tag{1.19}$$

Since the curl of the gradient of any scalar function vanishes identically, we may write

$$\vec{B} = \vec{\nabla}\,\psi \tag{1.20}$$

where ψ is a magnetic scalar potential. Substituting (1.20) into (1.14) and taking the divergence of both sides we obtain Laplace equation

$$\vec{\nabla}\cdot(\vec{\nabla}\psi) = \nabla^2\,\psi = 0 \tag{1.21}$$

since the divergence of the curl of any vector function vanishes identically. The only solution of this equation, which is not singular and finite at infinity, is ψ = constant, so that $\vec{B} = 0$. Hence, there is no magnetic field associated with these space charge oscillations. In summary, the electron plasma oscillations are *stationary*, *longitudinal* and *electrostatic*. They are also referred to as *Langmuir oscillations*. When the effect of the pressure gradient force is included in the equation of motion (1.3), complemented by an adiabatic energy equation,these oscillations become propagating disturbances commonly known as *space charge waves* or *Langmuir waves*. Characteristic values of the electron plasma frequency for various laboratory and cosmic plasmas are given in Fig. 2 of Chapter 1.

2. THE DEBYE SHIELDING PROBLEM

To examine the mechanism by which the plasma strives to shield its interior from a disturbing electric field, consider a plasma whose equilibrium state is perturbed by an electric field due to an external charged particle. For that matter, this electric field may also be considered to be due to one of the charged particles inside the plasma, isolated for observation. For definiteness, we assume this *test particle* to have a positive charge + Q, and choose a spherical coordinate system whose origin coincides with the position of the test particle. We are interested in determining the electrostatic potential $\phi(\vec{r})$ that is established near the test charge Q, due to the *combined* effects of the test charge and the distribution of charged particles surrounding it. Since the positive test charge Q attracts the negatively charged particles and repels the positively charged ones, the number densities of the electrons $n_e(\vec{r})$ and of the ions $n_i(\vec{r})$ will be slightly different near the origin (test particle), whereas at large distances from the origin the electrostatic potential vanhishes so that $n_e = n_i = n_o$. Since this is a steady state problem under the action of a conservative electric field, we have

$$\vec{E}(\vec{r}) = - \vec{\nabla}\, \phi(\vec{r}) \tag{2.1}$$

and from (7.5.16) it follows that

$$n_e(\vec{r}) = n_0 \exp\,[\,e\, \phi(\vec{r})/kT\,] \tag{2.2}$$

$$n_i(\vec{r}) = n_0 \exp\,[\,-\,e\, \phi(\vec{r})/kT\,] \tag{2.3}$$

where we have assumed that the electrons and ions (of charge e) have the same temperature T.

The total electric charge density $\rho(\vec{r})$, including the test charge Q, can be expressed as

$$\rho(\vec{r}) = -\, e\, [n_e(\vec{r}) - n_i(\vec{r})] + Q\, \delta(\vec{r}) \tag{2.4}$$

where $\delta(\vec{r})$ denotes the Dirac delta function. Using (2.2) and (2.3),

$$\rho(\vec{r}) = -\, en_0 \{ \exp[\, e\phi(\vec{r})/kT\,] - \exp[\,-\,e\, \phi(\vec{r})/kT\,] \} + Q\, \delta(\vec{r}) \tag{2.5}$$

Substituting (2.1) and (2.5) into the following Maxwell equation

$$\vec{\nabla} \cdot \vec{E}(\vec{r}) = \rho(\vec{r})/\varepsilon_0 \tag{2.6}$$

gives the differential equation

$$\nabla^2\, \phi(\vec{r}) - \frac{en_0}{\varepsilon_0} \left\{ \exp\,[\,e\, \phi(\vec{r})/kT\,] - \exp\,[\,-\,e\, \phi(\vec{r})/kT\,] \right\} = - \frac{Q}{\varepsilon_0}\, \delta(\vec{r}) \tag{2.7}$$

which permits the evaluation of the electrostatic potential $\phi(\vec{r})$.

We assume now that the perturbing electrostatic potential is weak so that the electrostatic potential energy is much less than the mean thermal energy, that is,

$$e\, \phi(\vec{r}) << kT \tag{2.8}$$

Under this condition we can use the approximation

$$\exp [\pm e\, \phi(\vec{r})/kT] \cong 1 \pm e\, \phi(\vec{r})/kT \tag{2.9}$$

Therefore, (2.7) simplifies to

$$\nabla^2 \phi(\vec{r}) - (2/\lambda_D^2)\, \phi(\vec{r}) = -(Q/\varepsilon_0)\, \delta(\vec{r}) \tag{2.10}$$

where λ_D denotes the Debye length

$$\lambda_D = (\varepsilon_0 kT/n_0 e^2)^{1/2} = (1/\omega_{pe})(kT_e/m_e)^{1/2} \tag{2.11}$$

Since the problem has spherical symmetry, the electrostatic potential depends only on the radial distance r measured from the position of the test particle, being independent of the spatial orientation of $\vec{r}$. Thus, using spherical coordinates, (2.10) can be written (for $r \neq 0$) as

$$\left(\frac{1}{r^2}\right) \frac{d}{dr}\left[r^2 \frac{d}{dr}\, \phi(r)\right] - \frac{2}{\lambda_D^2}\, \phi(r) = 0; \quad (r \neq 0) \tag{2.12}$$

In order to solve this equation we note initially that for an isolated particle of charge + Q, in free space, the electric field is directed radially outward and is given by

$$\vec{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}\, \hat{r} \tag{2.13}$$

so that the electrostatic Coulomb potential $\phi_c(r)$ due to this isolated charged particle in free space is

$$\phi_c(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r} \tag{2.14}$$

In the very close proximity of the test particle the electrostatic potential should be the same as that for an isolated particle in free space. Hence, it is appropriate to seek the solution of (2.12) in the form

$$\phi(r) = \phi_c(r)\, F(r) = \frac{Q}{4\pi\varepsilon_0} \frac{F(r)}{r} \tag{2.15}$$

where the function $F(r)$ must be such that $F(r) \rightarrow 1$ when $r \rightarrow 0$. Furthermore, the electrostatic potential $\phi(r)$ is required to vanish at infinity, that is, $\phi(r) \rightarrow 0$ when $r \rightarrow \infty$. Substituting (2.15) into (2.12) yields the following differential equation for $F(r)$

$$d^2 F(r)/dr^2 = (2/\lambda_D^2)\, F(r) \tag{2.16}$$

This simple differential equation for $F(r)$ has the solution

$$F(r) = A \exp\,(\sqrt{2}\, r/\lambda_D) + B \exp\,(-\sqrt{2}\, r/\lambda_D) \tag{2.17}$$

The condition that $\phi(r)$ vanishes for large values of r requires $A = 0$. Also, the condition that $F(r)$ tends to one when r tends to zero requires $B = 1$. Therefore, the solution of (2.12) is

$$\phi(r) = \phi_C(r) \exp\,(-\sqrt{2}\, r/\lambda_D) = \frac{1}{4\pi\varepsilon_0}\, \frac{Q}{r} \exp\,(-\sqrt{2}\, r/\lambda_D) \tag{2.18}$$

This result is commonly known as the *Debye potential*, since this nonrigorous derivation was first presented by Debye and Hückel in their theory of electrolytes. It shows that $\phi(r)$ becomes much less than the ordinary Coulomb potential once r exceeds the distance λ_D, called the *Debye length* (see Fig. 2). Hence, we can say in a crude way that a charged particle in a plasma interacts effectively only with particles situated at distances less than one Debye length away, and it has a negligible influence on particles lying at distances greater than one Debye length.

The charge Q of the test particle is neutralized by the charge distribution surrounding the test particle. From (2.5) and (2.9) we obtain for the charge density

$$\rho(\vec{r}) = -\, 2 n_0 e^2\, \phi(\vec{r})/kT + Q\, \delta(\vec{r}) \tag{2.19}$$

Substituting $\phi(\vec{r})$ by the Debye potential (2.18), we obtain

$$\rho(\vec{r}) = -\, (Q/2\pi r \lambda_D^2) \exp\,(-\sqrt{2}\, r/\lambda_D) + Q\, \delta(\vec{r}) \tag{2.20}$$

To obtain the total charge we integrate (2.20) over all space,

$$q_t = \iiint \rho(\vec{r})\, d^3r = -\frac{Q}{2\pi\, \lambda_D^2} \int_0^\infty \frac{1}{r} \exp\left(-\sqrt{2}\, r/\lambda_D\right) 4\pi r^2\, dr +$$

$$+ Q \iiint \delta(\vec{r})\, d^3r = 0 \qquad (2.21)$$

since the first integral gives $-Q$, whereas the second one is equal to $+Q$. The principal contribution to the first integral in (2.21) comes from the plasma particles lying in the very close neighborhood of the test particle, since the integrand falls off exponentially with increasing values of r. Thus, the neutralization of the test particle takes place effectively on account of the charged particles inside the Debye sphere. From (2.2) and (2.3) we see that in the neighborhood of the test particle the electron number density is larger than the ion number density, on account of the fact that the positive test particle attracts the electrons and repels the ions. Therefore, in the close proximity of the test particle there is an imbalance of charge and, consequently, an electric field. We have seen that the

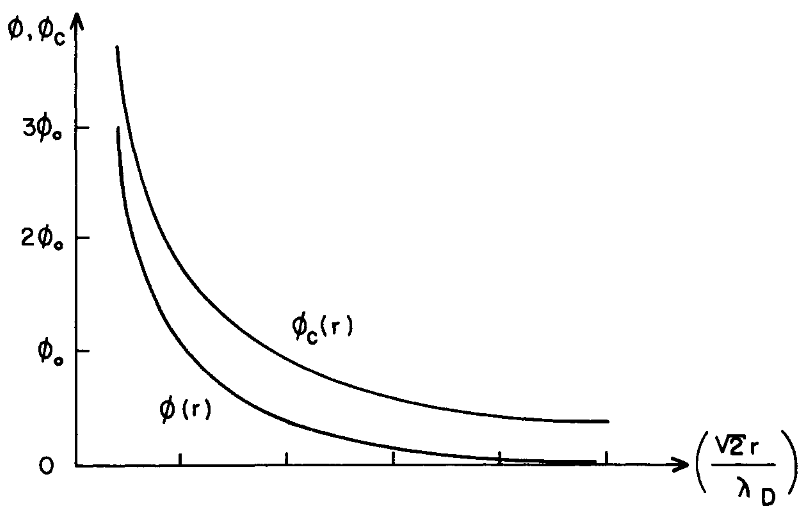

Fig. 2 - Electrostatic Coulomb potential $\phi_c(r)$ and Debye potential $\phi(r)$ as a function of distance r from the test charge Q. ($\phi_o = \sqrt{2}\, Q/4\pi\varepsilon_o\lambda_D$).

shielding of this electric field is effectively completed over a distance of the order of λ_D. Thus, for a plasma to be considered macroscopically neutral, it is necessary that its typical characteristic dimension L be much greater than λ_D. This quantitative *criterion* for the definition of a plasma has been previously discussed in Chapter 1.

An important point to be noted in the result (2.18) is that, for $r \to 0$, the Debye potential becomes very large and the assumption $e\phi(r) << kT$ is unlikely to be fulfilled. To verify the validity of this approximation, and consequently of (2.18), note that using (2.18) with $Q = e$ we have

$$\frac{e\phi}{kT} = \frac{e^2 \exp(-\sqrt{2}\, r/\lambda_D)}{4\pi\varepsilon_0 rkT} = \frac{\lambda_D}{3N_D} \frac{\exp(-\sqrt{2}\, r/\lambda_D)}{r} \tag{2.22}$$

where N_D is the number of electrons inside a Debye sphere. Since N_D is very large for virtually all plasmas, it is evident that the ratio given in (2.23) is much less than one, except when r is less than λ_D/N_D. Therefore, the Debye potential (2.18) is consistent with the approximation $e\phi << kT$ used to derive it, if we restrict attention to distances from the test particle greater than λ_D/N_D.

As a final point, we note that in the derivation of the Debye potential which appears extensively in the literature, it is usual to ignore ion motion and to assume a constant ion number density equal to the unperturbed electron number density. In this case the factor of 2 disappears from (2.10), and the expression for the Debye potential becomes

$$\phi(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r} \exp(-r/\lambda_D) \tag{2.23}$$

3. DEBYE SHIELDING USING THE VLASOV EQUATION

In this section we analyze the Debye potential problem from the point of view of kinetic theory. As before, we suppose that a test charge + Q is placed at the origin of a spherical coordinate system inside the plasma. In order to determine the steady state electron and ion distribution functions, f_e and f_i, and the electrostatic potential $\phi(\vec{r})$ near the test charge, let us consider the steady state Vlasov equations for the electrons and the ions, with only the electric field $\vec{E} = -\vec{\nabla}\phi$ in the Lorentz force term,

$$\vec{v} \cdot \vec{\nabla} f_e + (e/m_e)\ (\vec{\nabla}\phi) \cdot \vec{\nabla}_v\ f_e = 0 \tag{3.1}$$

$$\vec{v} \cdot \vec{\nabla} f_i - (e/m_i)\ (\vec{\nabla}\phi) \cdot \vec{\nabla}_v\ f_i = 0 \tag{3.2}$$

Since

$$n_\alpha(\vec{r}) = \int_v f_\alpha(\vec{r},v)\ d^3v \tag{3.3}$$

the total charge density (including the test particle) can be expressed as

$$\rho(\vec{r}) = -e \int_v (f_e - f_i)\ d^3v + Q\ \delta(\vec{r}) \tag{3.4}$$

and Poisson equation for this case becomes

$$\nabla^2\phi - \frac{e}{\varepsilon_o} \int_v (f_e - f_i)\ d^3v = - \frac{Q}{\varepsilon_o}\ \delta(\vec{r}) \tag{3.5}$$

Eqs. (3.1), (3.2) and (3.5) constitute three equations to be solved simultaneously to determine f_e, f_i and ϕ.

The solution of the Vlasov equations (3.1) and (3.2) can be expressed in terms of the Maxwellian distribution function and the Boltzmann factor (see section 5, Chapter 7), as

$$f_\alpha(\vec{r},v) = f_{o\alpha}(v)\ \exp\left[-q_\alpha\ \phi(\vec{r})/kT\right] \tag{3.6}$$

When the electrostatic potential vanishes, this distribution function goes into the Maxwellian distribution $f_{o\alpha}(v)$, with zero drift velocity. Substituting (3.6), with $\alpha = e, i$, into (3.5), yields

$$\nabla^2\phi - \frac{e}{\varepsilon_o} \left[\exp\ (e\,\phi/kT) \int_v f_{oe}(v)\ d^3v - \exp\ (-\,e\,\phi/kT) \int_v f_{oi}(v)\ d^3v\right]$$

$$= - \frac{Q}{\varepsilon_o}\ \delta(\vec{r}) \tag{3.7}$$

Denoting the electron and ion number densities under equilibrium conditions (when ϕ vanishes) by n_0,

$$n_0 = \int_v f_{0\alpha}(v)\, d^3v \qquad \alpha = e, i \tag{3.8}$$

(3.7) becomes

$$\nabla^2\phi - \frac{en_0}{\varepsilon_0}\left[\exp(e\phi/kT) - \exp(-e\phi/kT)\right] = -\frac{Q}{\varepsilon_0}\,\delta(\vec{r}) \tag{3.9}$$

This equation is identical to (2.7), yielding the same result as before for the Debye potential.

4. PLASMA SHEATH

When a material body is immersed in a plasma, the body acquires a net negative charge and therefore a negative potential with respect to the plasma potential. In the region near the wall of the body there is a boundary layer, known as the *plasma sheath*, in which the electron and ion number densities are different. Inside the plasma sheath the potential increases monotonically from a negative value on the wall to the value corresponding to the unperturbed plasma. The thickness of the plasma sheath, where departures from macroscopic electric neutrality occur, is found to be of the order of a Debye length.

A satisfactory mathematical treatment of this phenomenon is quite involved and cannot be presented here. However, the basic physical factors responsible for the formation of the plasma sheath are fairly simple to understand. In this section we will set up the problem mathematically and obtain some approximate results. The problem is strongly dependent on the particular geometry under consideration. For simplicity, we shall assume that the wall bounding the plasma is an infinite plane surface at $x = 0$, with the plasma in the region $x > 0$, and that there are no variations with respect to the coordinates y and z.

4.1 Physical mechanism

We begin with a descriptive account of the physical mechanism responsible for the formation of the plasma sheath. The charged particles in the plasma that strike the wall in virtue of their random thermal motions are for the most part lost to the plasma. The ions generally recombine at the wall and return to the plasma as neutral particles, whereas the electrons may either recombine there or they may enter the conduction band if the surface is a metal. We have seen in section 4, of Chapter 7, that the *random* particle flux, that is, the number of particles that hit the surface per unit area and per unit time, from one side only, for the case of an isotropic velocity distribution function, is given by [see Eqs. (7.4.35) and (7.4.18)]

$$\Gamma_\alpha = n_\alpha \langle v \rangle_\alpha / 4 \tag{4.1}$$

where $\langle v \rangle_\alpha$ is the average particle speed for the α species. For the Maxwell-Boltzmann velocity distribution function we have found that [see Eq. (7.4.20)]

$$\langle v \rangle_\alpha = (8/\pi)^{1/2} \, (kT_\alpha/m_\alpha)^{1/2} \tag{4.2}$$

so that a typical value for the random particle flux in this case is

$$\Gamma_\alpha = n_\alpha \, (kT_\alpha/2\pi m_\alpha)^{1/2} \tag{4.3}$$

It is evident from this result that if initially the electron and ion number densities are equal, then the random particle flux for the electrons, Γ_e, greatly exceeds that for the ions, Γ_i, since in general $(T_e/m_e)^{1/2}$ is much larger than $(T_i/m_i)^{1/2}$. For the least heavy ion, hydrogen, for example, $m_i/m_e = 1836$. Therefore, the wall in contact with the plasma rapidly accumulates a negative charge, since initially more electrons reach the wall than positive ions. This negative potential repels the electrons and attracts the ions so that the electron flux diminishes and the ion flux increases. Eventually, the negative potential at the wall becomes large enough in magnitude to equalize the rate at which electrons and ions hit the surface. At this floating negative potential the wall and the plasma reach a dynamical equilibrium such that the net current at the wall is zero.

4.2 Electric potential on the wall

To estimate the value of the potential on the wall after the plasma sheath has been established, consider a steady state situation and let the electric potential $\phi(x)$ at the wall $(x = 0)$ be given by

$$\phi(0) = \phi_W \tag{4.4}$$

Let us choose the reference potential inside the plasma, at a very large distance from the wall, equal to zero,

$$\phi(\infty) = 0 \tag{4.5}$$

The electrons and the ions are assumed to be in thermodynamic equilibrium at the same temperature T, under the action of the conservative electric field associated with the negative potential on the wall. At $x \rightarrow \infty$ the plasma is unperturbed and the electron and ion number densities are each equal to n_o. According to the results of section 5, Chapter 7, the electron and ion number densities can be expressed as

$$n_e(\vec{r}) = n_o \exp [e\,\phi(\vec{r})/kT] \tag{4.6}$$

$$n_i(\vec{r}) = n_o \exp [-e\,\phi(\vec{r})/kT] \tag{4.7}$$

It is important to note at this point that (4.6) and (4.7) do not take into account the particle drift velocity towards the wall. Since the electrons and the ions impinging on the wall surface are for the most part lost to the plasma, there must be a steady flux of both species towards the wall to replenish this charge particle loss. Despite this inadequacy, (4.6) and (4.7) will still be used to obtain an approximate expression for the potential on the wall and afterwards, in order to study the inner structure of the plasma sheath, we will take into account the particle drift in an approximate manner using the hydrodynamic equations.

One of the boundary conditions of the problem is that under equilibrium conditions there must be no charge build up at the wall $(x = 0)$, so that

$$J_e(0) = J_i(0) \tag{4.8}$$

Using (4.3), (4.6) and (4.7), considering singly charged ions,

$$(1/m_e)^{1/2} \exp\,(e\,\phi_w/kT) = (1/m_i)^{1/2} \exp\,(-\,e\,\phi_w/kT) \tag{4.9}$$

which may be written as

$$\exp\,(-\,2\,e\,\phi_w/kT) \;=\; (m_i/m_e)^{1/2} \tag{4.10}$$

Taking the natural logarithm of both sides, and solving for the wall potential, we obtain

$$\phi_w = -\,(kT/4e)\,\ell n(m_i/m_e) \tag{4.11}$$

Other more accurate methods of calculating the wall potential yield results which, for $T_e = T_i$, agree qualitatively with the one given in (4.11), despite the inadequacy of (4.6) and (4.7) which neglect the particle drift velocity towards the wall.

Note from (4.11) that the magnitude of the potential energy near the wall $|e\,\phi_w|$ is of the same order as the average thermal energy kT of the particles in the plasma, since

$$|\,e\,\phi_w\,|/kT \;=\; (1/4)\,\ell n(m_i/m_e) \tag{4.12}$$

For a hydrogen ion, for example, $|\,e\,\phi_w\,|/kT$ is approximately equal to 2, whereas for heavier ions it may be close to 3.

4.3 Inner structure of the plasma sheath

To investigate the inner structure of the plasma sheath consider the equations of conservation of particles and momentum for the electrons and ions under steady state conditions, with spatial dependence only on the x direction. The equation of conservation of particles becomes

$$d(n_\alpha u_\alpha)/dx = n_\alpha(du_\alpha/dx) + u_\alpha(dn_\alpha/dx) = 0 \tag{4.13}$$

In the momentum conservation equation we neglect viscosity effects and approximate the kinetic pressure dyad by a scalar pressure. The ideal gas equation of state, $p_\alpha = n_\alpha k T_\alpha$, can be used to introduce the temperature, which is assumed to be constant. Collisions are neglected, since the thickness of the plasma sheath is much less than the mean free path for the plasma particles. With this assumptions and in the absence of a magnetic field, the equation of motion becomes (with $\vec{E} = -\vec{\nabla}\phi$, and $D/DT = \partial/\partial t + \vec{u}_\alpha \cdot \vec{\nabla} = u_\alpha\, d/dx$)

$$m_\alpha u_\alpha (du_\alpha/dx) = -(kT_\alpha/n_\alpha)(dn_\alpha/dx) - q_\alpha\, d\phi/dx \tag{4.14}$$

In order to simplify the analysis we shall make two approximations. From (4.13), we can write

$$dn_\alpha/dx = -(n_\alpha/u_\alpha)(du_\alpha/dx) \tag{4.15}$$

and the ratio of the magnitude of the term in the left-hand side of (4.14) to the first term in the right-hand side can be expressed as

$$\frac{|(m_\alpha u_\alpha)(du_\alpha/dx)|}{|(kT/n_\alpha)\,(dn_\alpha/dx)|} = \frac{m_\alpha\, u_\alpha^2}{kT} \tag{4.16}$$

The two approximations consist in neglecting the left-hand side term of (4.14) for the electrons, whereas for the ions we neglect the first term in the right-hand side of (4.14). Explicitly, we take for the electrons (neglecting electron inertia)

$$(kT_e/n_e)\, dn_e/dx - e\, d\phi/dx = 0 \tag{4.17}$$

and for the ions (assuming cold ions)

$$m_i\, u_i\, du_i/dx + e\, d\phi/dx = 0 \tag{4.18}$$

These two approximations are justified only if the thermal energy of the *electrons* is much *larger* than their kinetic energy, and if the thermal energy of the *ions* is much *smaller* than their kinetic energy. Thus, we require from (4.15) that

$$m_e\, u_e^2 \ll kT \ll m_i\, u_i^2 \tag{4.19}$$

in order to justify the approximations in (4.17) and (4.18). We shall assume that the condition (4.19) is satisfied in the plasma sheath but it remains to be justified later in this section.

If we integrate (4.17), we obtain

$$e\,\phi(x) = kT\,\ell n\, n_e(x) + (\text{constant}) \tag{4.20}$$

and using the condition that $n_e = n_o$ when $\phi = 0$, we find

$$n_e(x) = n_o \exp\,[e\,\phi(x)/kT] \tag{4.21}$$

This result is identical to (4.6), which is not surprising, since the condition $m_e\, u_e^2 \ll kT$ implies in neglecting the electron inertia $(m_e = 0)$ and consequently their kinetic energy.

For the ions, we first integrate (4.13) to find

$$n_i(x)\, u_i(x) = C_1 \tag{4.22}$$

and then integrate (4.18) to obtain

$$e\,\phi(x) + (m_i/2)\, u_i^2(x) = C_2 \tag{4.23}$$

where C_1 and C_2 are constants. The boundary conditions require that at $x = \infty$ we must have $\phi(\infty) = 0$, $n_i(\infty) = n_o$ and $u_i(\infty) = u_{oi}$. Thus,

$$C_1 = n_o\, u_{oi} \quad ; \quad C_2 = m_i\, u_{oi}^2/2 \tag{4.24}$$

and using these results in (4.22) and (4.23), we find

$$n_i(x)\, u_i(x) = n_o\, u_{oi} \tag{4.25}$$

$$e\,\phi(x) + (m_i/2)\, u_i^2(x) = m_i\, u_{oi}^2/2 \tag{4.26}$$

These two equations can be combined to eliminate $u_i(x)$ and solved for $n_i(x)$, giving

$$n_i(x) = n_o \, [1 - 2e\,\phi(x)/(m_i\, u_{oi}^2)]^{-1/2} \tag{4.27}$$

This expression for $n_i(x)$ is substantially different from the one given in (4.7) and this difference is due to the importance of the ion drift velocity. We now find that, since $\phi(x) < 0$ in the sheath, $n_i(x)$ decreases slowly towards the wall rather than increasing as predicted by (4.7). Physically, this behavior is due to the fact that the negative potential on the wall causes $u_i(x)$ to increase as the ions approach the wall and since the flux $n_i(x)\; u_i(x)$ must stay constant, in virtue of (4.25), it turns out that $n_i(x)$ must decrease according to (4.27). This behavior is illustrated schematically in Fig. 3.

To obtain the differential equation satisfied by the electrostatic potential $\phi(x)$, we substitute (4.21) and (4.27) into Poisson equation

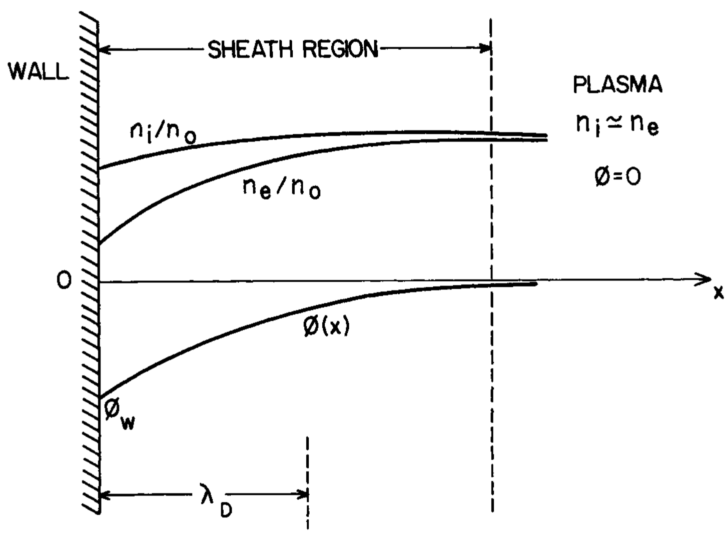

Fig. 3 - Diagram showing the variation of the electrostatic potential $\phi(x)$ and the number densities n_e and n_i inside the plasma sheath near an infinite plane wall.

$$\nabla^2 \phi = e\,(n_e - n_i)/\varepsilon_o \tag{4.28}$$

to obtain

$$\frac{d^2\phi}{dx^2} = \frac{n_o e}{\varepsilon_o}\left[\exp\left(\frac{e\phi}{kT}\right) - \left(1 - \frac{2e\phi}{m_i u_{oi}^2}\right)^{-1/2}\right] \tag{4.29}$$

In this equation, the drift velocity u_{oi}, far away from the wall, still needs to be determined. This equation is nonlinear and in order to facilitate its analytical solution we need to make one more approximation. Since we have seen that $|e\phi|$ ranges from zero in the plasma to a value of the order of kT on the wall and since we have also assumed that $m_i u_{oi}^2$ is larger than kT, we will restrict our attention to the region near the plasma edge of the sheath and assume further that $|e\phi|$ is small compared to both kT and $m_i u_{oi}^2$. Thus, in the region near the edge of the sheath adjacent to the plasma, we can expand the terms in the right-hand side of (4.29), for $(e\phi/kT) \ll 1$ and $(e\phi/m_i u_{oi}^2) \ll 1$, as

$$\exp(e\phi/kT) \cong 1 + (e\phi/kT) \tag{4.30}$$

$$[1 - 2e\phi/(m_i u_{oi}^2)]^{-1/2} \cong 1 + e\phi/(m_i u_{oi}^2) \tag{4.31}$$

with the result that (4.29) reduces to

$$d^2\phi/dx^2 \cong \phi/X^2 \tag{4.32}$$

where

$$X^2 = \lambda_D^2\,[1 - kT/(m_i u_{oi}^2)]^{-1} \tag{4.33}$$

The solution of (4.32), with the boundary condition $\phi(\infty) = 0$, is

$$\phi(x) = A\exp(-x/X) \tag{4.34}$$

where A is a constant. Since we have assumed that $kT \ll m_i u_{oi}^2$, it follows from (4.33) that X is real and approximately equal to λ_D.

Therefore, we find that $\phi(x)$ increases exponentially (since A must be negative) as we move from inside the sheath into the plasma and goes to zero at very large distances from the wall. Since $X \cong \lambda_D$, this increase effectively takes place within a distance of the order of a Debye length. This solution for $\phi(x)$ is strictly valid only near the plasma edge of the sheath, but if it is continued to apply throughout the plasma sheath, we can impose the boundary condition on the wall, that is, $\phi(0) = \phi_w$, which requires that $A = \phi_w$.

If kT were greater than $m_i u_{oi}^2$, than X would be imaginary and the electric potential would be an oscillating function of distance near the wall. The condition

$$kT < m_i u_{oi}^2 \tag{4.35}$$

must therefore be satisfied for the formation of a plasma sheath. It is known as the *Bohm criterion.*

It is not a trivial matter to determine the potential on the wall using the hydrodynamic equations. All the approximate methods that have been suggested give results which, for $T_e = T_i$, agree reasonably well with the approximate value for ϕ_w given in (4.11). Furthermore, there is no consistent way to determine the ion drift velocity u_{oi} at $x = \infty$, but an approximate estimate can be obtained as follows. Since the ion flux must be constant, from (4.25) we can equate the ion flux $n_o u_{oi}$ at $x = \infty$ to its value $\Gamma_i(0)$ at the wall. Using (4.3) with $n_i(\vec{r})$ as given by (4.7) evaluated at the wall, we find

$$u_{oi} = (kT/2\pi m_i)^{1/2} \exp\left(-e\phi_w/kT\right) \tag{4.36}$$

Similarly, for the electrons we can equate the electron flux $n_o u_{oe}$ at $x = \infty$ to its value $\Gamma_e(0)$ at the wall. From (4.3), with $n_e(\vec{r})$ given by (4.6), evaluated at the wall, we find

$$u_{oe} = (kT/2\pi m_e)^{1/2} \exp\left(e\phi_w/kT\right) \tag{4.37}$$

and using (4.9) we see that

$$u_{oe} = u_{oi} \tag{4.38}$$

To verify the validity of the approximations indicated in (4.19), we note first that from the mass conservation equation the particle flux $n_\alpha(x)\ u_\alpha(x)$ must be constant for all x and equal to the value $n_o u_{o\alpha}$. From (4.21) we see that the *minimum* value of $n_e(x)$ is $n_o \exp(e\phi_w/kT)$ since ϕ_w is negative. Therefore,

$$u_e = (n_o/n_e)\ u_{oe} \leqq u_{oe} \exp(-e\,\phi_w/kT) \tag{4.39}$$

and using (4.37), we get

$$u_e \leqq (kT/2\pi m_e)^{1/2} \tag{4.40}$$

or

$$kT/m_e\ u_e^2 \geqq 2\pi \tag{4.41}$$

in agreement with the assumption (4.19). Similarly, from (4.27) the *maximum* value of $n_i(x)$ is n_o, so that

$$u_i = (n_o/n_i)\ u_{oi} \geqq u_{oi} \tag{4.42}$$

and using (4.36), we obtain

$$u_i \geqq (kT/2\pi m_i)^{1/2} \exp(-e\,\phi_w/kT) \tag{4.43}$$

or

$$kT/m_i\ u_i^2 \leqq 2\pi \exp(2e\,\phi_w/kT) \cong 0.1 \tag{4.44}$$

where the result in the right has been obtained substituting ϕ_w by the value given in (4.11). Therefore, in view of (4.41) and (4.44), the thermal energy of the electrons is seen to be greater than their kinetic energy, whereas for the ions the opposite situation is verified. From (4.44) we can verify that the Bohm criterion, for the formation of the plasma sheath, is also satisfied.

Although the quantitative aspects of the discussion presented here are very approximate, it provides, however, a satisfactory qualitative picture of the sheath.

5. THE PLASMA PROBE

The plasma probe is a device that has been widely used to measure the temperature and density of the plasma particles, both in the laboratory and in space vehicles. The electrostatic probe was developed by Langmuir and Mott-Smith, and the physical mechanism of its operation can be well explained using the theory of plasma sheaths presented in the previous section. A conducting probe, or electrode, is immersed in a plasma and the current that flows through it is measured for various potentials applied to the probe. The temperature and number density of the electrons can then be obtained from the characteristics of the resulting current-potential curve. When the surface of the probe is plane, the current-potential curve has a shape like that illustrated in Fig. 4.

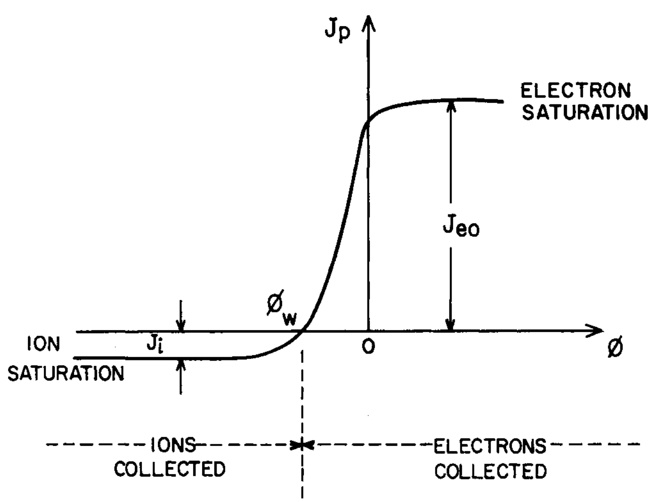

Fig. 4 - Characteristics of the current-potential curve of an electrostatic probe immersed in a plasma. The floating potential of the probe with reference to the plasma potential is denoted by ϕ_w.

The probe, when inserted in the plasma, is surrounded by a plasma sheath which shields the major portion of the plasma from the disturbing probe field. The thickness of the sheath is of the order of a Debye length. When no current flows through the electrode, it is at the negative floating potential ϕ_W, which is the wall potential discussed in the previous section. Under these conditions, the number of electrons reaching the probe per unit time is equal to the number of positive ions reaching the probe per unit time. We assume the current to be positive when it flows in the direction away from the probe. The current associated with the electrons is directed away from the probe and therefore is considered positive. Consequently, the electric current associated with the flow of ions is negative. Under equilibrium conditions, there is no net current flowing through the probe and its potential is the floating potential ϕ_W. When the potential is made more negative than ϕ_W, the electron current is reduced due to the increased repulsive force imposed by the probe electric field on the electrons. As the potential is made more negative, the contribution to the electric current arising from the electrons will eventually become neglibible and the total electric current asymptotically approaches a constant negative value, corresponding to the electric current density J_i associated with only the flow of ions. The ions that reach the edge of the plasma sheath fall into the potential well and their current is practically unaffected if the potential is made even more negative. On the other hand, when the probe potential is increased from the negative value ϕ_W, more electrons reach the probe than ions per unit time due to the decrease in the repulsion force on the electrons and the net electric current becomes positive. When the electric potential is zero, that is, when the probe is at the same potential as the plasma, there is no electric field near the electrode and, since the average thermal velocity of the electrons is much greater than that of the ions, the electron current density J_{eo} (for $\phi = 0$) is much greater than the ion current density. If the potential is made sufficiently positive, a situation arises in which the current associated with the ions becomes negligible, but all the electrons that reach the edge of the sheath are collected by the probe. The electron current density reaches a fairly constant value for sufficiently high positive values of ϕ. This plateau region in the probe current-potential curve is called the region of *saturation* of the electron current. For higher positive values of ϕ there are complications in the current-potential characteristic of the electrode due to the occurrence of another phenomena.

An approximate expression for the magnitude of the electron current density, away from the region of saturation, can be obtained from (4.6) as

$$J_e = J_{eo} \exp (e \phi / kT_e) \tag{5.1}$$

where J_{eo} is the electron current density when the electric potential is zero. Since for $\phi = 0$ we have $\Gamma_e = n_e < v >_e/4$, and using (4.2) for the average electron speed, we obtain,

$$J_{eo} = en_e \, (kT_e/2\pi m_e)^{1/2} \tag{5.2}$$

where n_e is the electron number density in the unperturbed plasma region. Note that when ϕ is negative the ions reaching the edge of the sheath continue to fall into the negative potential of the probe and hence the ion current density is a constant (J_i) in the negative potential region. Thus, we can express the probe current density, in the region where $\phi < 0$, as

$$J_p = J_{eo} \exp (e \phi / kT_e) - J_i \qquad (\text{for } \phi < 0) \tag{5.3}$$

From this equation we can deduce the result

$$T_e = \frac{e}{k} \left\{ \frac{d}{d\phi} \left[\ln (J_p + J_i) \right] \right\}^{-1} \tag{5.4}$$

This expression can be used to determine the electron temperature as follows. First, the electrode is made sufficiently negative with reference to the plasma potential, so that the current that flows through the probe is due to the ions only. The measurement of this current gives directly the value J_i. Then, the current-potential characteristic of the probe is measured and a plot of $\ln (J_p + J_i)$ as a function of ϕ is made. This curve has a straight-line section corresponding to the probe potential less than the plasma potential, and the slope of this straight line gives the value of

$$\frac{d}{d\phi} \left[\ln (J_p + J_i) \right] \tag{5.5}$$

which, when substituted in (5.4), gives the electron temperature in the plasma.

After the electron temperature T_e has been determined, we can evaluate the electron number density from (5.2), which can be written as,

$$n_e = (J_{eo}/e)(2\pi m_e/kT_e)^{1/2} \tag{5.6}$$

The value of J_{eo} is determined by measuring the probe current corresponding to the plateau (electron saturation) region of the current-potential characteristic of the probe.

PROBLEMS

11.1 - Consider a stationary plasma (electrons and one type of ions) under steady state conditions at a uniform temperature T_o, when perturbed by a point charge + Q placed at the origin of a coordinate system. Using the collisionless hydrodynamic equation for the electrons and ions

$$\rho_{m\alpha}\, D\vec{u}_\alpha/Dt = n_\alpha\, q_\alpha\, \vec{E} - \vec{\nabla} p_\alpha \quad ; \quad (\alpha = e,i)$$

with the ideal gas law $p_\alpha = n_\alpha k\, T_o$, and Poisson equation

$$\nabla^2\phi(\vec{r}) = -\rho(\vec{r})/\varepsilon_o$$

obtain the following differential equation

$$\nabla^2\phi(\vec{r}) - (2/\lambda_D^2)\,\phi(\vec{r}) = -(Q/\varepsilon_o)\,\delta(\vec{r})$$

for the Debye potential $\phi(\vec{r})$. Assume that the number density of each species can be expressed as $n_\alpha = n_o + n'_\alpha$, where n_o is constant and $|n'_\alpha| \ll n_o$. What are the approximations necessary to obtain this result?

11.2 - Analyze the Debye potential problem considering only the motion of the electrons (ions stay immobile), and show that in this case the differential equation for the electric potential $\phi(\vec{r})$ is

$$\nabla^2\phi(\vec{r}) - (1/\lambda_D^2)\ \phi(\vec{r}) = -\ (Q/\varepsilon_o)\ \delta(\vec{r})$$

11.3 - When the macroscopic neutrality of a plasma is instantaneously perturbed by external means, the electrons react in a such a way as to give rise to oscillations at the electron plasma frequency $\omega_{pe} = (n_o e^2/m_e \varepsilon_o)^{1/2}$. Consider these oscillations, but including the motion of the ions. Show that in this case the natural frequency of oscillation of the net charge density is given by

$$\omega = (\omega_{pe}^2 + \omega_{pi}^2)^{1/2}.$$

where $\omega_{pi} = (n_o e^2/m_i \varepsilon_o)^{1/2}$. Use the linearized equations of continuity and of momentum for each species, and Poisson equation, considering only the electric force due to the internal charge separation.

11.4 - Evaluate the negative electrostatic potential ϕ_w which appears on an infinite plane wall immersed in a plasma consisting of electrons of charge - e and ions of charge Ze, under steady state conditions. Denote the electron and ion temperatures by T_e and T_i, respectively.

11.5 - Deduce an expression for the Debye potential for a test particle of charge + Q immersed in a plasma consisting of electrons (charge - e) and ions of charge Ze, the temperature of the electrons and the ions being T_e and T_i, respectively. Show that, if $T_e \gg T_i$, then the Debye length is governed by the ion temperature T_i.

11.6 - Using the following expressions for the electron and ion number densities

$$n_e(\vec{r}) = n_o \exp [e\,\phi(\vec{r})/kT]$$

$$n_i(\vec{r}) = n_o \exp [-e\ \phi(\vec{r})/kT)]$$

in the plasma sheath region formed between an infinite plane and a semi-infinite plasma, deduce the differential equation satisfied by the electric potencial $\phi(x)$ in the plasma sheath. Show that this differential equation can be written in the form

$$d^2F/d\zeta^2 = \sinh(F)$$

where $F = e\phi/kT$ and $\zeta = \sqrt{2}\,x/\lambda_D$. Assuming that at $x = \infty$ we have $n_e = n_i = n_o$, $F = 0$ and $dF/d\zeta = 0$, show that

$$F(\zeta) = 4 \tanh^{-1} \{ \exp[-(\zeta - \zeta_o)] \}$$

where ζ_o is a constant. Denoting the potential at the wall by ϕ_w and assuming that $e\phi/kT \ll 1$, show that

$$\phi(x) = \phi_w \exp(\sqrt{2}\,x/\lambda_D)$$

with

$$\phi_w = (4kT/e) \exp(\zeta_o)$$

11.7 - For the plasma sheath region formed in the vicinity of a plane wall immersed in a plasma, assume that the ions at the plasma edge of the sheath can be described by a shifted Maxwellian distribution function

$$f_i(v) = n_o \left(\frac{m_i}{2\pi kT} \right)^{3/2} \exp\left[- \frac{m_i\,(\vec{v} - \vec{u}_o)^2}{2\,kT} \right]$$

with drift velocity $\vec{u}_o = u_o\,\hat{x}$. Prove that the ion flux Γ_{ix}, at the edge of the sheath, is given by

$$\Gamma_{ix} = n_o\,(kT/2\pi m_i)^{1/2}\,\{ \exp(-y^2) + y\sqrt{\pi}\,[1 + \mathrm{erf}(y)] \}$$

where $y = u_o(m_i/2kT)^{1/2}$ and $\mathrm{erf}(y)$ is the error function, defined by

$$\mathrm{erf}(y) = \frac{2}{\sqrt{\pi}} \int_0^y \exp(-s^2)\,ds$$

Calculate $d\Gamma_{ix}/dy$. Note that the error function vanishes for $y = 0$, increases monotonically as y increases and tends asymptotically to unity as

$y \rightarrow \infty$. Also note that

$$\frac{d}{dy}\,[\,\mathrm{erf}(y)\,] = \frac{2}{\sqrt{\pi}}\,\exp\,(-y^2)$$

11.8 - From an experimental current-potential curve of a Langmuir probe of area A immersed in a plasma, such as shown in Fig. P11.1, where the electric potentials are measured with respect to a fixed reference potential, explain how you can determine J_i, J_{eo}, the space potential ϕ_s, the floating potential ϕ_f with respect to ϕ_s (note that $\phi_f - \phi_s = \phi_w$), T_e and n_e.

11.9 - The Langmuir plasma probe has been widely used in satellites to measure space plasma properties. In one valuable technique, circuits are arranged that measure directly $dI_p/d\phi$ and $d^2I_p/d\phi^2$, where $I_p = J_p\,A$ and A is the probe's area. Use (5.3) to show that

$$\frac{(dI_p/d\phi)}{(d^2I_p/d\phi^2)} = \frac{kT_e}{e}$$

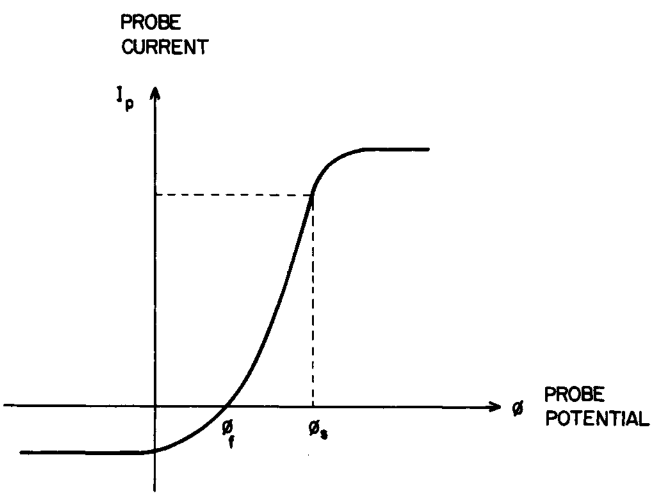

Fig. P11.1

which gives directly the electron temperature T_e. Next show that J_{eo} can be calculated from T_e and $dI_p/d\phi$ at a known value of ϕ, according to

$$J_{eo} = (kT_e/Ae) \exp(-e\phi/kT_e)(dI_p/d\phi)$$

The electron density n can then be calculated from (5.6).

11.10 - An electron gas (Lorentz gas) in a background of stationary ions, is acted upon by a weak, externally applied electric field $\vec{E}$, under steady state conditions. Using the Boltzmann equation for the electrons, with the relaxation model for the collision term (with a constant collision frequency ν),

$$(\delta f/\delta t)_{coll} = -\nu(f - f_o)$$

and considering the adiabatic case for which

$$n(\vec{r})\,[T(\vec{r})]^{-3/2} = \text{constant}$$

show that the electron distribution function is given by

$$f = f_o \left\{ 1 - \frac{1}{\nu} \left[\left(\frac{mv^2}{2kT} \right) \left(\vec{v} \cdot \frac{\vec{\nabla} T}{T} \right) + \left(\frac{e}{kT} \right) (\vec{v} \cdot \vec{E}) \right] \right\}$$

Assume that $f = f_o + f_1$, where $|f_1| << f_o$ and where f_o is the following modified Maxwellian distribution

$$f_o(\vec{r},v) = n(\vec{r}) \left[\frac{m}{2\pi k T(\vec{r})} \right]^{3/2} \exp\left[-\frac{mv^2}{2kT(\vec{r})} \right]$$

and neglect all second-order terms in the Boltzmann equation. Consider the term involving $\vec{\nabla} f_1$ as a second-order quantity.

11.11 - Using the distribution function of the previous problem, evaluate the electric current density $\vec{J}$ to show that the presence of a temperature gradient gives rise to an electric current associated with thermoelectric effects.

11.12 - Consider problem 11.10, but taking $\vec{E} = 0$ and, instead of the adiabatic case, consider a constant kinetic pressure

$$p = n(\vec{r})\, k\, T(\vec{r}) = \text{constant}$$

(a) Show that the electron distribution function is given by

$$f = f_o \left\{ 1 - \frac{1}{\nu} \left[-\frac{5}{2} + \left(\frac{mv^2}{2kT} \right) \right] \vec{v} \cdot \frac{\vec{\nabla} T}{T} \right\}$$

(b) Evaluate the heat flux vector $\vec{q}$ and show that it can be written as

$$\vec{q} = - K \vec{\nabla} T(\vec{r})$$

where the thermal conductivity K is $5kp/2m\nu$.

(c) What is the value of the electric current density $\vec{J}$ in this case?

11.13 - In the previous problem, consider that n = constant and that f_o is the following modified Maxwell-Boltzmann distribution function

$$f_o(\vec{r},v) = n \left[\frac{m}{2\pi kT(\vec{r})} \right]^{3/2} \exp \left[-\frac{mv^2}{2kT(\vec{r})} \right]$$

Calculate the electron distribution function $f(\vec{r},\vec{v})$ and show that the heat flux vector is given by

$$\vec{q} = - K\, \vec{\nabla}\, T(\vec{r})$$

Determine the expression for the thermal conductivity K.

11.14 - In Problem 11.12, inculde the presence of an external magnetostatic field $\vec{B}$ in the z direction and deduce the following expression for the nonequilibrium distribution function:

$$f = f_o \left\{ 1 - \frac{1}{\nu} \left(-\frac{5}{2} + \frac{mv^2}{2kT} \right) \left[\frac{\nu\omega_{ce}}{(\nu^2 + \omega_{ce}^2)}\, v_x \left(\hat{x} \frac{\nu}{\omega_{ce}} - \hat{y} \right) + \right. \right.$$

$$+ \frac{\nu\omega_{ce}}{(\nu^2 + \omega_{ce}^2)} v_y \left(\hat{x} + \hat{y} \frac{\nu}{\omega_{ce}} \right) + v_z \hat{z} \Big] \cdot \frac{\vec{\nabla} T}{T} \Big\}$$

Show that the heat flux vector $\vec{q}$ can be expressed as

$$\vec{q} = - \bar{\bar{K}} \cdot \vec{\nabla} T(\vec{r})$$

where $\bar{\bar{K}}$ is the dyadic thermal conductivity, which in matrix form is given by

$$\bar{\bar{K}} = \begin{pmatrix} K_\perp & -K_H & 0 \\ K_H & K_\perp & 0 \\ 0 & 0 & K_\parallel \end{pmatrix}$$

with

$$K_\perp = \frac{\nu^2}{(\nu^2 + \omega_{ce}^2)} K_o$$

$$K_H = \frac{\nu\omega_{ce}}{(\nu^2 + \omega_{ce}^2)} K_o$$

$$K_\parallel = (5kp/2m\nu) \equiv K_o$$

Note: The solution of the differential equation

$$df_1/d\phi + (\nu/\omega_{ce}) f_1 = - (1/\omega_{ce}) \vec{v} \cdot \vec{\nabla} f_o$$

is given by

$$f_1 = - \exp\left(- \frac{\nu}{\omega_{ce}} \phi \right) \frac{1}{\omega_{ce}} \int_{-\infty}^{\phi} d\phi' \exp\left(\frac{\nu}{\omega_{ce}} \phi' \right) \vec{v} \cdot \vec{\nabla} f_o$$

11.15 - The *coefficient of viscosity* η is defined as the shear stress produced by unit velocity gradient. For the p_{xz} component of the kinetic pressure dyad we have

$$p_{xz} = - \eta \frac{d}{dz} u_x(z)$$

Assume the following form for the equilibrium velocity distribution function of the electrons

$$f_o(\vec{r},\vec{v}) = n \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left[- \frac{m}{2kT} \left\{ [v_x - u_x(z)]^2 + v_y^2 + v_z^2 \right\} \right]$$

which indicates the presence of an average velocity $u_x(z)$ in the x direction with a gradient in the z direction. In the absence of external forces and using the relaxation model for the collision term with a constant collision frequency ν, let $f = f_o + f_1$ with $|f_1| << f_o$ in the Boltzmann equation, under steady state conditions, to show that

$$f_1(\vec{r},\vec{v}) = - \frac{2}{\nu} \left(\frac{m}{2kT}\right) f_o(\vec{r},\vec{v})\, v_z [v_x - u_x(z)] \frac{\partial}{\partial z} u_x(z)$$

Next, calculate p_{xz} and show that the coefficient of viscosity is given by $\eta = nkT/\nu$.